



NORTH SYDNEY BOYS HIGH SCHOOL

2013 YEAR 12 ASSESSMENT TASK 4

Mathematics

General Instructions

- Working time – ~~60~~ minutes
- **Reading time – 5 minutes**
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher: Please colour

- ☐ Mr Lowe
- ☐ Ms Juhn
- ☐ Mr Lam
- ☐ Mr Berry
- ☐ Mr Lin

Student Number: _____

(To be used by the exam markers only.)

Question No	1-2	3	4	5	6	7	8	Total	Total
Mark	$\overline{6}$	$\overline{4}$	$\overline{11}$	$\overline{6}$	$\overline{9}$	$\overline{6}$	$\overline{8}$	$\overline{50}$	$\overline{100}$

1. A box has 5 Psychology books and 3 Astronomy books in it.
Two books are selected at random from the box.
 - (a) Draw a probability tree to show all the possible outcomes. 2
 - (b) Find the probability that :
 - (i) the two books are Astronomy books. 1
 - (ii) at least one of the books is a Psychology book. 2

2. In a game of Yahtzee, 5 standard dice are rolled.
Find the probability of rolling all the same number. 1

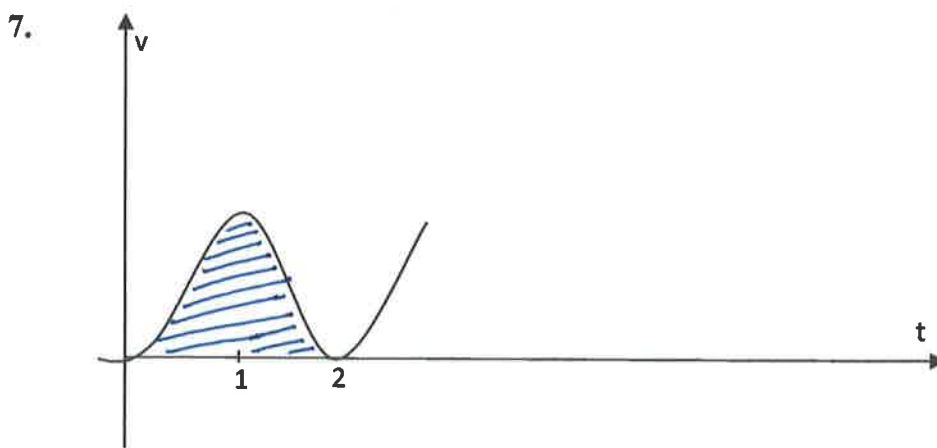
3. David and Ryan play a game where they each take turns at throwing two ordinary dice. The winner is the first person to throw a double.
David throws first.
 - (a) Show that the probability that Ryan wins the game on his first throw is $\frac{5}{36}$. 2
 - (b) Show that the probability Ryan wins the game on his first or second throw is given by $\frac{5}{36} + \frac{5^3}{6^4}$. 2

4. A super ball is thrown into the air and its path of motion is described by the equation $x = 24t - 3t^2$. Time is measured in seconds and distance travelled in metres.
 - (a) At what times is the displacement zero? 2
 - (b) Find the maximum height reached by the ball. 2
 - (c) Find the velocity at $t = 1$ 2
 - (d) Find the average velocity in the first 5 seconds 2
 - (e) Find the average speed in the first 5 seconds. 2

exam continues overleaf.....→

5. A particle moves in a straight line. At time t seconds, its displacement, x metres from a fixed point O on the line is given by
- $$x = 1 - \cos \pi t$$
- (a) What is the initial displacement of the particle? 1
- (b) Find an expression for the velocity of the particle at any time t . 1
- (c) What is the velocity of the particle at time $t = \frac{1}{6}$? 1
- (d) Find an expression for the acceleration of the particle at any time t . 1
- (e) At what time does the particle **first** reach its maximum speed? 2

6. In a fish hatchery, the fish population, N , satisfies the equation $N = N_0 e^{kt}$ where N_0 and k are constants and t is measured in months.
- (a) Initially there were 1000 fish in the hatchery and at the end of 5 months there were 5000. Find the value of k correct to 3 decimal places. 2
- (b) Find the number of fish in the hatchery at the end of 8 months. (Give your answer correct to the nearest hundred.) 2
- (c) At the end of which month will the fish population exceed 50 000 for the first time? 3
- (d) At what rate is the population increasing at the end of six months? (Give your answer correct to the nearest hundred fish per month.) 2

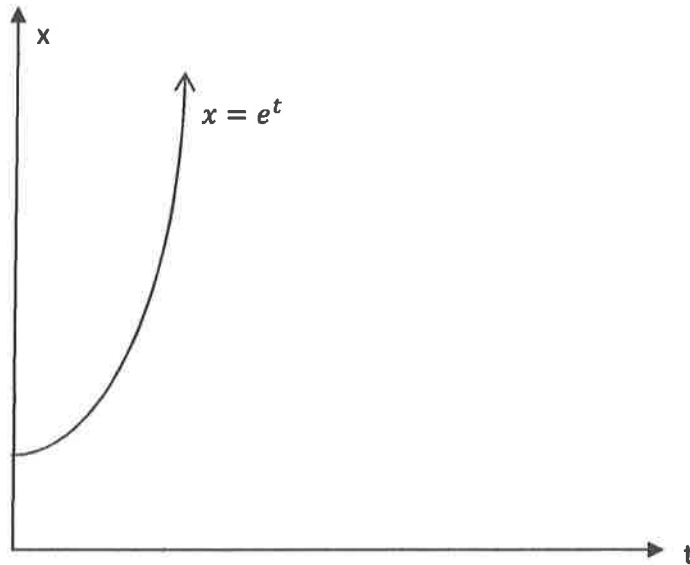


The velocity - time graph of a particle starting from rest at the origin is shown in the diagram.

- (a) When is the particle at rest again? 1
- (b) Does the particle change its direction during the motion? Give reasons. 2
- (c) Give the first time that the acceleration of the particle is zero. 1
- (d) Give a physical interpretation of the shaded area bounded by the t axis and the curve. 1
- (e) Sketch the displacement-time graph of the particle. 2

exam continues overleaf.....→

8. Two particles P and Q are moving along a horizontal line. At any time t seconds, the position of the particle P is given by $x = e^t$ and the position of particle Q is given by $x = 1 + 6e^{-t}$. The diagram below shows the position of particle P, x metres, at any time t seconds.



- | | |
|---|---|
| (a) As time increases indefinitely, what position does the particle Q approach? | 1 |
| (b) On a pair of coordinate axes, sketch the path of particle Q. | 2 |
| (c) Calculate the position where the two particles meet. | 4 |
| (d) Explain why P and Q will never travel at the same velocity. | 1 |

END OF EXAMINATION

2 UNIT SOLUTIONS - ASSESS. TSK 4 2013

1. a)

1st

2nd

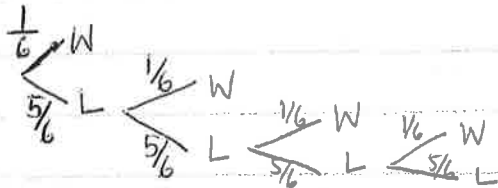
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graph LR; Root(( )) ---|5/8| P1[P]; Root ---|3/8| A1[A]; P1 ---|4/7| P2[P]; P1 ---|3/7| A2[A]; A1 ---|5/7| P3[P]; A1 ---|2/7| A3[A];
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$$b) i) P(A) = \frac{3}{8} \times \frac{2}{7}$$
$$= \frac{3}{28}$$

$$\begin{aligned} \text{(ii) } P(\text{at least one } P) &= 1 - P(AA) \\ &= 1 - \frac{3}{28} \\ &= \frac{25}{28} \end{aligned}$$

$$2. P(\text{same number}) = \left(\frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6}\right) \times 6$$
$$= \frac{1}{6^4}$$

3. D R D R



$$a) P(\text{Ryan wins}) = \frac{5}{6} \times \frac{1}{6} = \frac{5}{36}$$

$$b) P(\text{Ryan wins}) = \frac{5}{36} + \left(\frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} \right)$$
$$= \frac{5}{36} + \frac{5^3}{6^4}$$

④ $x = 24t - 3t^2$

a) $0 = 24t - 3t^2$

$0 = 3t(8-t)$

$t=0, t=8.$

b) Max height : $t=4$

$\therefore x = 24 \times 4 - 3 \times 16$

$= 48 \text{ m}$

OR Max height : $\dot{x} = 0.$

$0 = 24 - 6t$

$6t = 24$

$t = 4$

$\therefore x = 48 \text{ m}$

c) $\dot{x} = 24 - 6t$

When $t=1$

$\dot{x} = 24 - 6$

$\dot{x} = 18 \text{ m/s}$

⑤ $x = 1 - \cos \pi t$

a) When $t=0$

$x = 1 - \cos 0$

$x = 0$

b) $\dot{x} = \pi \sin \pi t$

c) When $t = \frac{1}{6}$

$\dot{x} = \pi \sin \frac{\pi}{6}$

$\dot{x} = \frac{\pi}{2} \text{ m/s}$

d) $\ddot{x} = \pi^2 \cos \pi t$

e) Max Speed : $\ddot{x} = 0$

$\pi^2 \cos \pi t = 0$

$\cos \pi t = 0$

$\pi t = \frac{\pi}{2}$

$t = \frac{1}{2} \text{ sec}$

d) Average Velocity = $\frac{x_2 - x_1}{t_2 - t_1}$

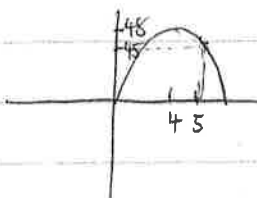
$= \frac{45 - 0}{5 - 0}$

$= 9 \text{ m/s}$

e) Average speed = $\frac{48 + 3}{5}$

$= \frac{51}{5} \text{ m/s}$

$= 10.2 \text{ m/s}$



6. $N = N_0 e^{kt}$

a) $N = 1000 e^{kt}$

When $t=5$, $N=5000$

$$5000 = 1000 e^{5k}$$

$$5 = e^{5k}$$

$$\ln 5 = 5k$$

$$k = \frac{1}{5} \ln 5$$

$$k = 0.322$$

b) When $t=8$.

$$N = 1000 e^{0.322 \times 8}$$

$$N = 13132.639 \text{ (using exact } k)$$

$$N = \underline{13100} \text{ (using exact } k)$$

$$N = 13144.1455 \text{ (using } 0.322 = k)$$

$$= \underline{13100}$$

c) $1000 e^{0.322t} > 50000$

$$e^{0.322t} > 50$$

$$0.322t > \ln 50$$

$$t > \frac{\ln 50}{0.322}$$

$$t > 12.149$$

* At the end of the
13th month

d) $\frac{dN}{dt} = k N_0 e^{kt}$

$$= 0.322 \times 1000 e^{0.322 \times 6}$$

$$= 2200 \text{ fish/month}$$

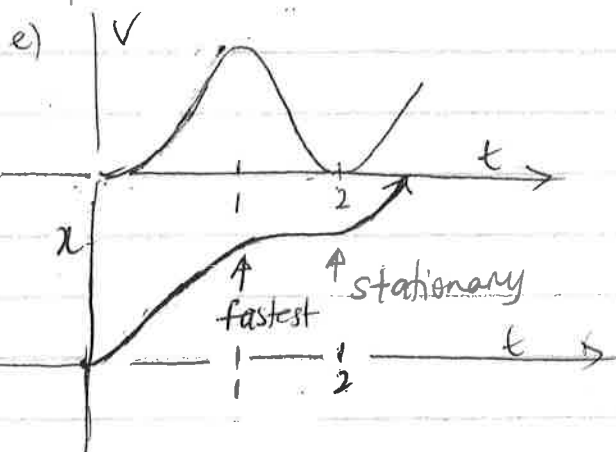
7,

a) At $t=1$

b) No as the velocity is always positive

c) $t=1$

d) distance travelled by the particle in the first 2 seconds.



8. P: $x = e^t$

Q: $x = 1 + 6e^{-t}$

a) As $t \rightarrow \infty$

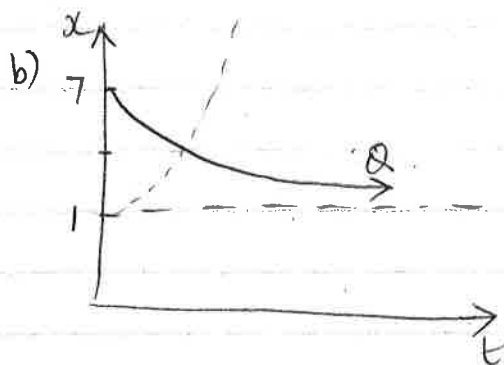
$$x = 1 + \frac{6}{e^t}$$

As $t \rightarrow \infty$

$$e^t \rightarrow \infty$$

$$\frac{6}{e^t} \rightarrow 0$$

$$\therefore x \rightarrow 1 \text{ m.}$$



c) Two particles meet at same x value.

$$1 + 6e^{-t} = e^t$$

$$1 + \frac{6}{e^t} = e^t$$

$$e^t + 6 = e^{2t}$$

$$e^{2t} - e^t - 6 = 0$$

$$\text{Let } a = e^t$$

$$\therefore a^2 - a - 6 = 0$$

$$(a-3)(a+2) = 0$$

$$a = 3 \text{ or } -2$$

$$\text{i.e. } e^t = 3 \text{ or } e^t = -2$$

$$t = \ln 3. \quad \text{no sol}$$

$$\text{i.e. } x = e^{\ln 3}$$

$$x = 3.$$

$\therefore$ They meet at $x = 3 \text{ m.}$

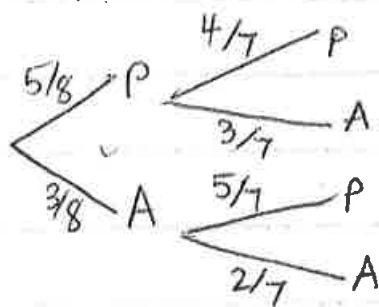
d) Never travel at

the same velocity
because ~~Q~~ is always
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velocity is negative.

P is always increasing
and hence velocity positive.

20 UNIT SOLUTIONS - ASSESS. TSK 4 2013

1. a) 1st 2nd

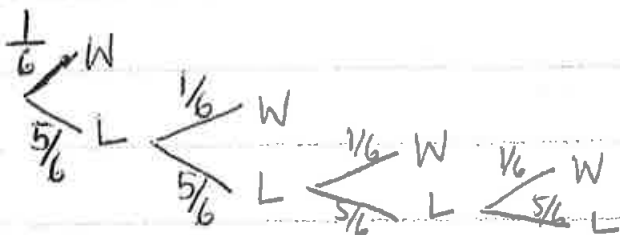


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$$d) \quad \ddot{x} = \pi^2 \cos \pi t$$

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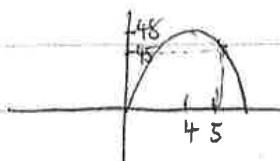
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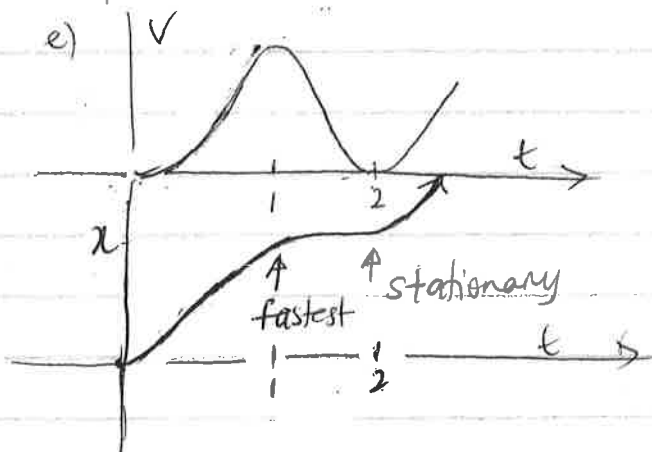
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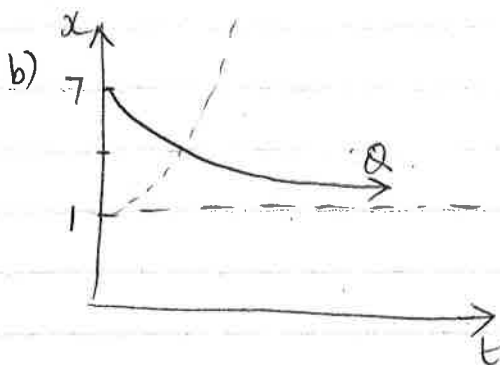
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the same velocity because ~~Q~~ is always decreasing and hence velocity is negative.

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